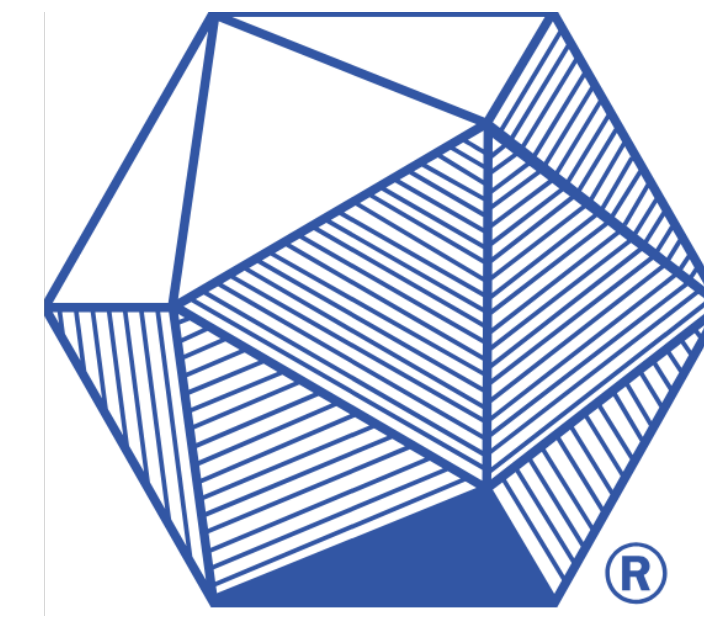




1. Motivation

We study the space of all 3-state birth–death Markov chains with nonzero transition probabilities.

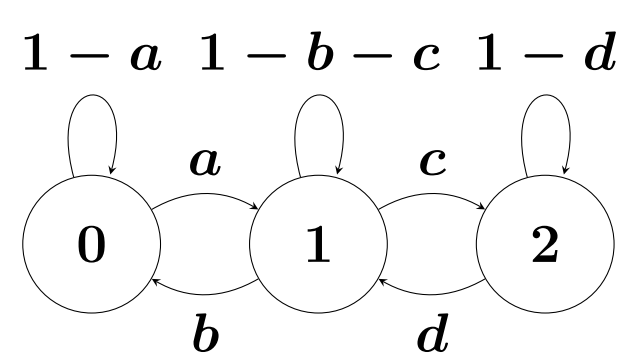
Goals:

- Understand how parameters determine stationary distributions.
- Partition chains by having the same steady state.
- Study the topology of the resulting quotient space.

Main conjecture:

The space of equivalence classes of 3-state birth–death chains with nonzero transition probabilities is homeomorphic to the 2-dimensional probability simplex without its boundary.

2. 3-State Birth–Death Chain



For a general 3-state birth–death chain, the transition matrix is

$$P = \begin{bmatrix} 1-a & a & 0 \\ c & 1-b-c & b \\ 0 & d & 1-d \end{bmatrix}$$

and the steady-state distribution is given by:

$$\pi(0) = \frac{cd}{cd+da+ab}, \quad \pi(1) = \frac{da}{cd+da+ab}, \quad \pi(2) = \frac{ab}{cd+da+ab}$$

Note that we assume $a, b, c, d > 0$, with $b + c < 1$, and that any steady-state in this space has all nonzero entries.

3. Equivalent Chains Condition

Consider two chains with transition probability matrices P_1 and P_2 , where

$$P_1 = \begin{bmatrix} 1-a_1 & a_1 & 0 \\ c_1 & 1-b_1-c_1 & b_1 \\ 0 & d_1 & 1-d_1 \end{bmatrix} \quad P_2 = \begin{bmatrix} 1-a_2 & a_2 & 0 \\ c_2 & 1-b_2-c_2 & b_2 \\ 0 & d_2 & 1-d_2 \end{bmatrix}$$

The two chains have the same steady state if and only if

$$[a_1, c_1] = k_1[a_2, c_2] \quad \text{and} \quad [b_1, d_1] = k_2[b_2, d_2]$$

where $a_2 k_1, c_2 k_1, b_2 k_2, d_2 k_2 \in (0, 1)$.

4. Stationary Distributions and Scalar Multiples

($\Rightarrow$) Assume M_1 and M_2 are two chains with the same steady state. Then

$$\begin{aligned} \pi(1)_1 &= \pi(1)_2 \\ \frac{d_1 a_1}{c_1 d_1 + d_1 a_1 + a_1 b_1} &= \frac{d_2 a_2}{c_2 d_2 + d_2 a_2 + a_2 b_2} \\ a_1 &= a_2 \cdot \frac{d_2(c_1 d_1 + d_1 a_1 + a_1 b_1)}{d_1(c_2 d_2 + d_2 a_2 + a_2 b_2)} \end{aligned}$$

Similarly, letting $\pi(0)_1 = \pi(0)_2$, we get

$$c_1 = c_2 \cdot \frac{d_2(c_1 d_1 + d_1 a_1 + a_1 b_1)}{d_1(c_2 d_2 + d_2 a_2 + a_2 b_2)}$$

Thus,

$$[a_1, c_1] = k_1[a_2, c_2]$$

and similarly

$$[b_1, d_1] = k_2[b_2, d_2].$$

($\Leftarrow$) Assume

$$[a_1, c_1] = k_1[a_2, c_2], \quad [b_1, d_1] = k_2[b_2, d_2].$$

Then

$$\pi(0)_1 = \frac{c_1 d_1}{c_1 d_1 + d_1 a_1 + a_1 b_1} = \frac{k_1 c_2 k_2 d_2}{k_1 k_2 (c_2 d_2 + d_2 a_2 + a_2 b_2)} = \pi(0)_2$$

Similarly,

$$\pi(1)_1 = \pi(1)_2, \quad \pi(2)_1 = \pi(2)_2.$$

5. Alpha and Beta Parameters

Additionally, two chains have the same steady state if and only if

$$P_1 = \begin{bmatrix} 1-a_1 & a_1 & 0 \\ \alpha a_1 & 1-b_1-\alpha a_1 & b_1 \\ 0 & \beta b_1 & 1-\beta b_1 \end{bmatrix} \quad P_2 = \begin{bmatrix} 1-a_2 & a_2 & 0 \\ \alpha a_2 & 1-b_2-\alpha a_2 & b_2 \\ 0 & \beta b_2 & 1-\beta b_2 \end{bmatrix}$$

where $\alpha a_1, \alpha a_2 \in (0, 1)$ and $\beta b_1, \beta b_2 \in (0, 1)$.

In other words, the steady state only depends on α and β . We have

$$(\alpha, \beta) \in (0, \infty)^2.$$

For the backward direction, we verify

$$\vec{\pi}_1 = \vec{\pi}_2 = \left[\frac{\alpha\beta}{\alpha\beta + \beta + 1}, \frac{\beta}{\alpha\beta + \beta + 1}, \frac{1}{\alpha\beta + \beta + 1} \right].$$

($\Rightarrow$) Assume both chains share steady state $\vec{\pi}$.

Claim: Any steady state can be written in this form by letting

$$\alpha = \frac{\pi(0)}{\pi(1)}, \quad \beta = \frac{\pi(1)}{\pi(2)}.$$

An arbitrary chain

$$P = \begin{bmatrix} 1-a & a & 0 \\ \alpha a & 1-b-\alpha a & b \\ 0 & \beta b & 1-\beta b \end{bmatrix}$$

has steady state $\vec{\pi}$.

6. Applying the Parameterization

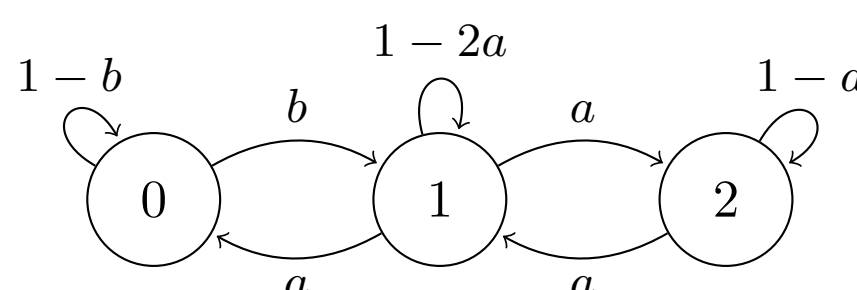
Thus, any chain with this steady state must take the form:

$$P_1 = \begin{bmatrix} 1-k_1 a & k_1 a & 0 \\ \alpha k_1 a & 1-k_2 b-\alpha k_1 a & k_2 b \\ 0 & \beta k_2 b & 1-\beta k_2 b \end{bmatrix} \quad P_2 = \begin{bmatrix} 1-k_3 a & k_3 a & 0 \\ \alpha k_3 a & 1-k_4 b-\alpha k_3 a & k_4 b \\ 0 & \beta k_4 b & 1-\beta k_4 b \end{bmatrix}$$

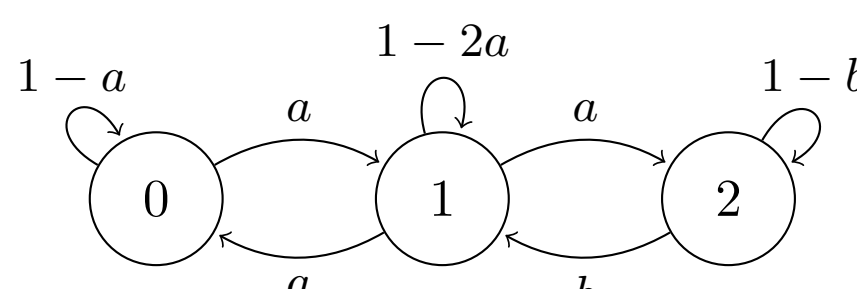
Let

$$a_1 = k_1 a, \quad b_1 = k_2 b, \quad a_2 = k_3 a, \quad b_2 = k_4 b.$$

7. Case 3 and 4: (b, a, a, a) and (a, a, a, b)



$$\begin{bmatrix} 1-b & b & 0 \\ a & 1-2a & a \\ 0 & a & 1-a \end{bmatrix}$$



$$\begin{bmatrix} 1-a & a & 0 \\ a & 1-2a & a \\ 0 & b & 1-b \end{bmatrix}$$

Eigenvalues: $\lambda_1 = 1$

$$\begin{aligned} \lambda_2 &= \frac{1}{2} \left(-\sqrt{5a^2 - 2ab + b^2} - 3a - b + 2 \right) \\ \lambda_3 &= \frac{1}{2} \left(\sqrt{5a^2 - 2ab + b^2} - 3a - b + 2 \right) \end{aligned}$$

Case 3

$$\pi(0) = \frac{a}{a+2b}$$

$$\pi(1) = \frac{b}{a+2b}$$

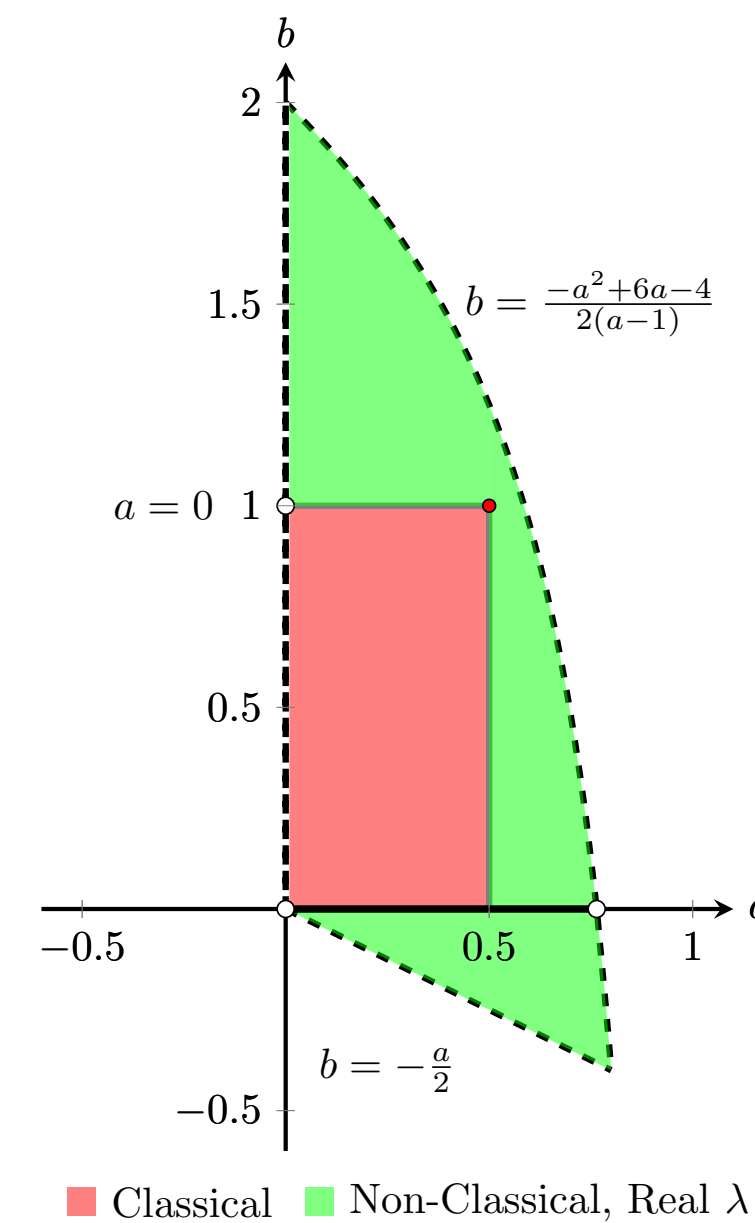
$$\pi(2) = \frac{b}{a+2b}$$

Case 4

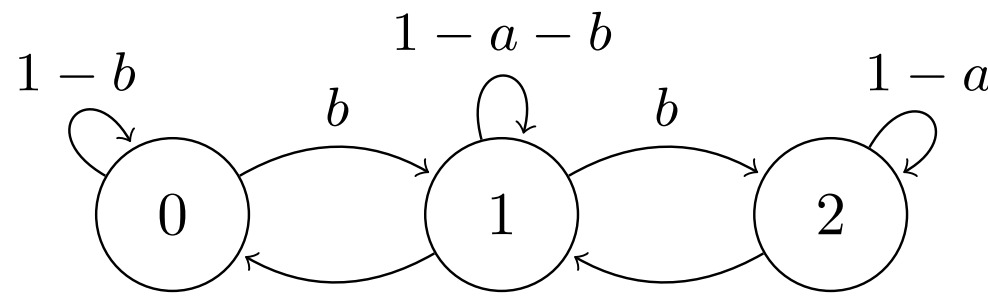
$$\pi(0) = \frac{b}{a+2b}$$

$$\pi(1) = \frac{b}{a+2b}$$

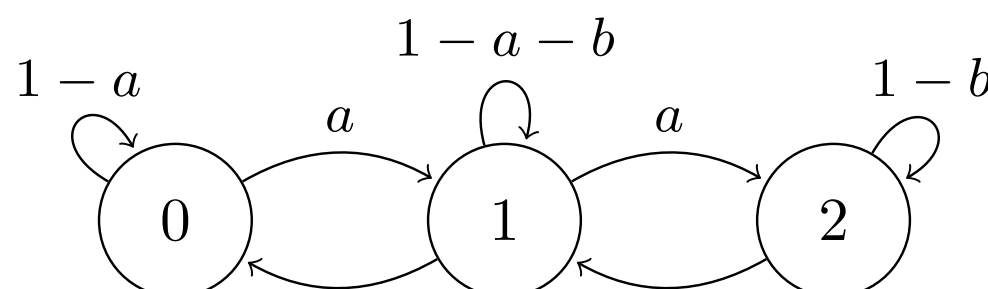
$$\pi(2) = \frac{a}{a+2b}$$



8. Case 5 and 6: (b, b, a, a) and (a, a, b, b)



$$\begin{bmatrix} 1-b & b & 0 \\ a & 1-a-b & b \\ 0 & a & 1-a \end{bmatrix}$$



$$\begin{bmatrix} 1-a & a & 0 \\ b & 1-a-b & a \\ 0 & b & 1-b \end{bmatrix}$$

Eigenvalues:

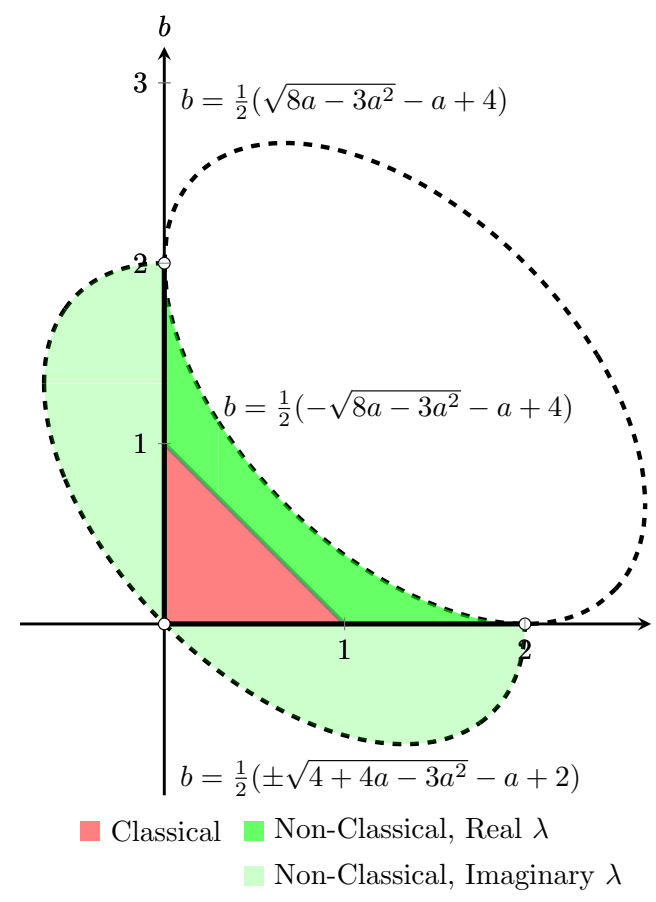
$$\lambda_1 = 1 \quad \lambda_2 = -\sqrt{ab} - a - b + 1 \quad \lambda_3 = \sqrt{ab} - a - b + 1$$

Case 5

$$\pi(0) = \frac{a^2}{a^2 + ab + b^2}$$

$$\pi(1) = \frac{ab}{a^2 + ab + b^2}$$

$$\pi(2) = \frac{b^2}{a^2 + ab + b^2}$$



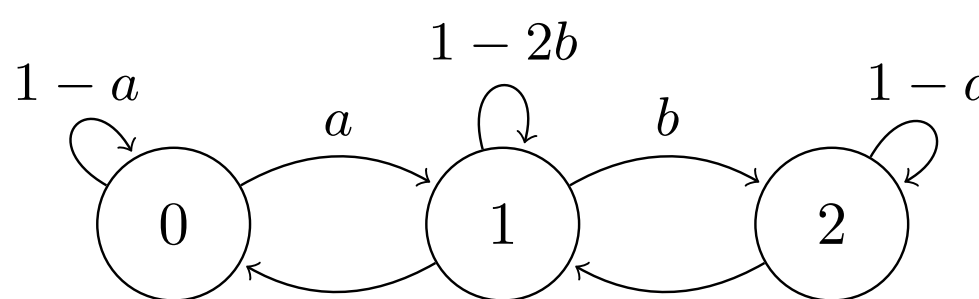
Case 6

$$\pi(0) = \frac{b^2}{a^2 + ab + b^2}$$

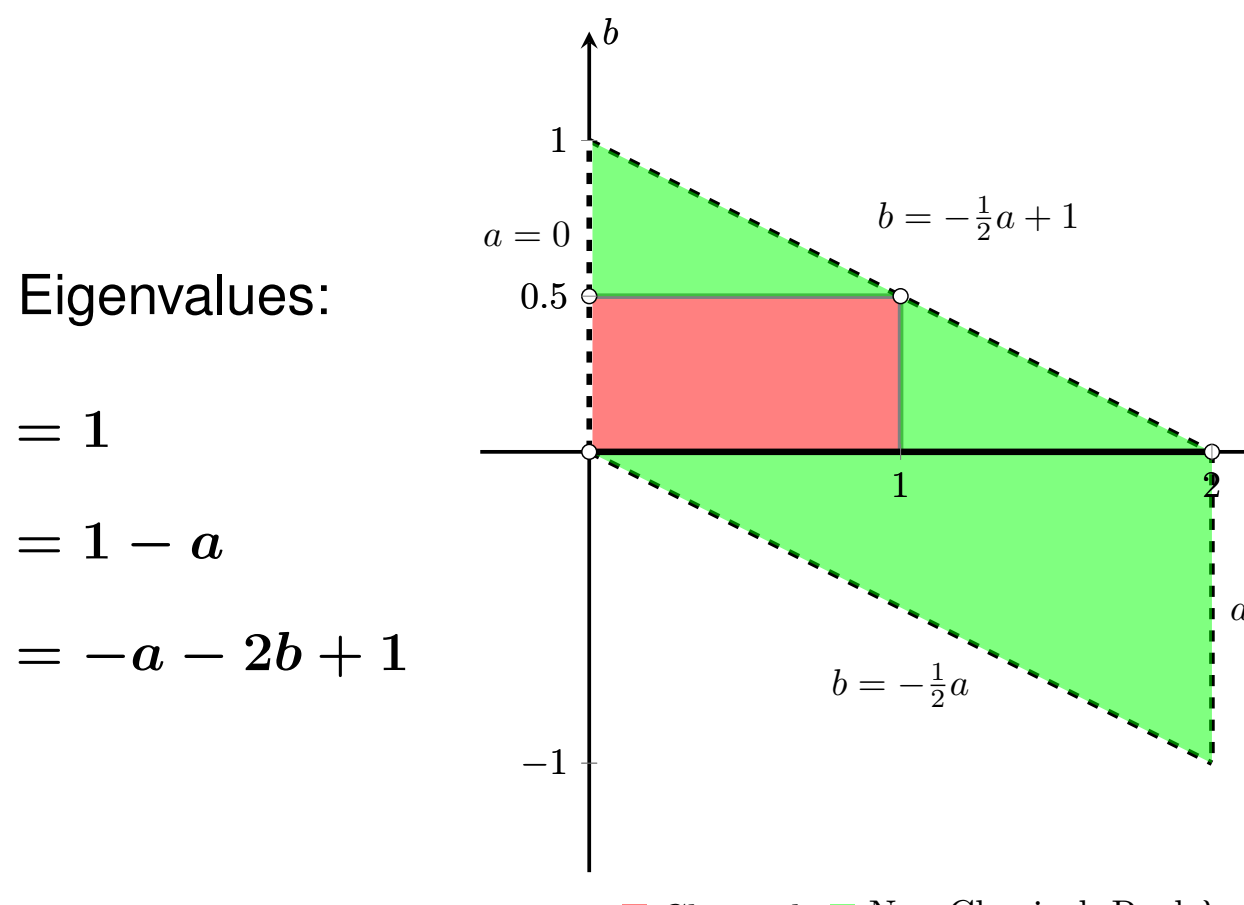
$$\pi(1) = \frac{ab}{a^2 + ab + b^2}$$

$$\pi(2) = \frac{a^2}{a^2 + ab + b^2}$$

9. Case 7: (a, b, b, a)



$$\begin{bmatrix} 1-a & a & 0 \\ b & 1-2b & b \\ 0 & a & 1-a \end{bmatrix}$$



Eigenvalues:

$$\lambda_1 = 1$$

$$\lambda_2 = 1 - a$$

$$\lambda_3 = -a - 2b + 1$$

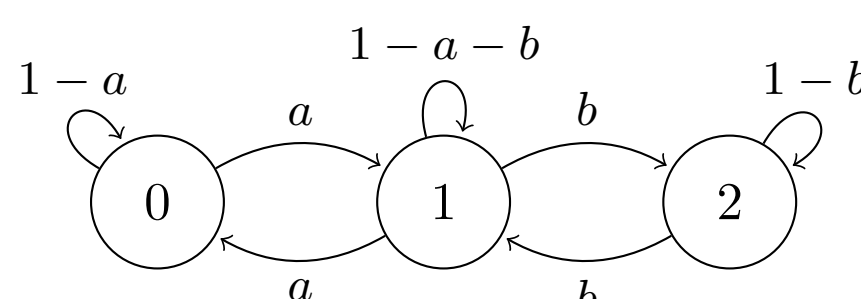
Case 7

$$\pi(0) = \frac{b}{a+2b}$$

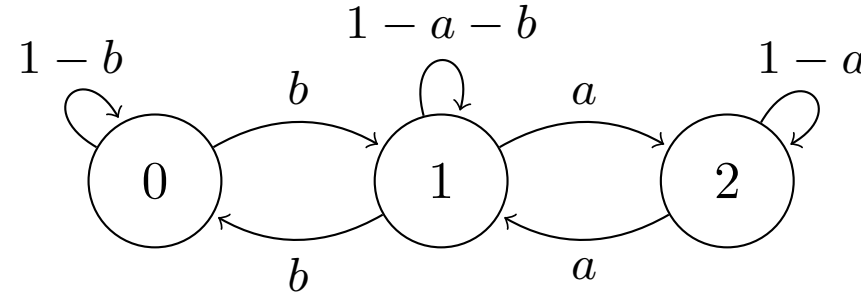
$$\pi(1) = \frac{a}{a+2b}$$

$$\pi(2) = \frac{b}{a+2b}$$

10. Case 8 and 9: (a, b, a, b) and (b, a, b, a)



$$\begin{bmatrix} 1-a & a & 0 \\ a & 1-a-b & b \\ 0 & b & 1-b \end{bmatrix}$$



$$\begin{bmatrix} 1-b & b & 0 \\ b & 1-a-b & a \\ 0 & a & 1-a \end{bmatrix}$$

Eigenvalues:

$$\lambda_1 = 1$$

$$\lambda_2 = -\sqrt{a^2 - ab + b^2} - a - b + 1$$

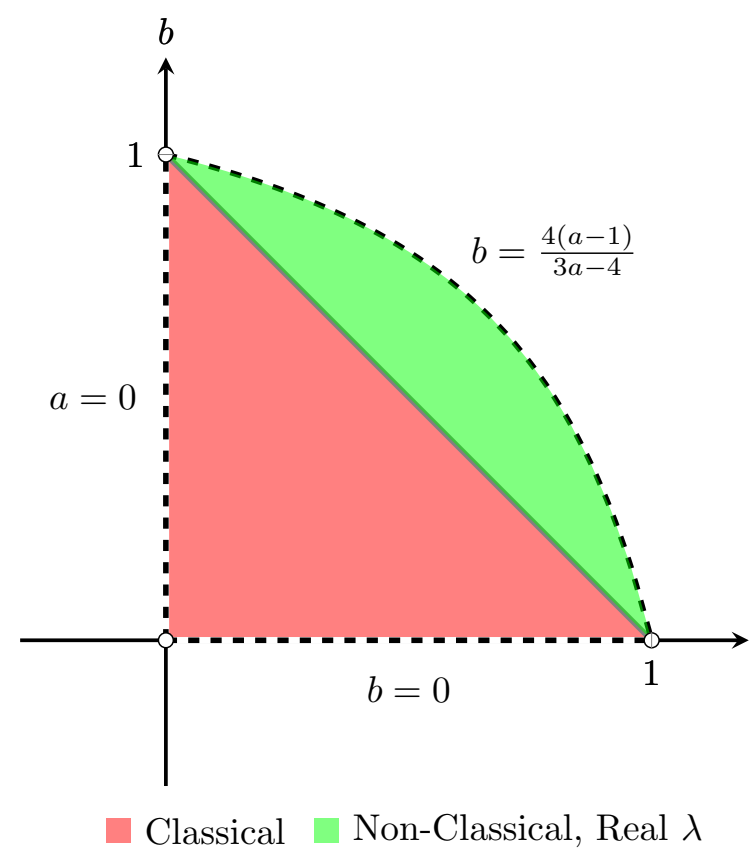
$$\lambda_3 = \sqrt{a^2 - ab + b^2} - a - b + 1$$

Case 8 and 9

$$\pi(0) = \frac{1}{3}$$

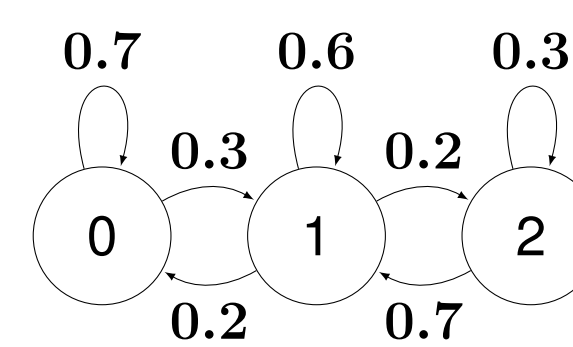
$$\pi(1) = \frac{1}{3}$$

$$\pi(2) = \frac{1}{3}$$

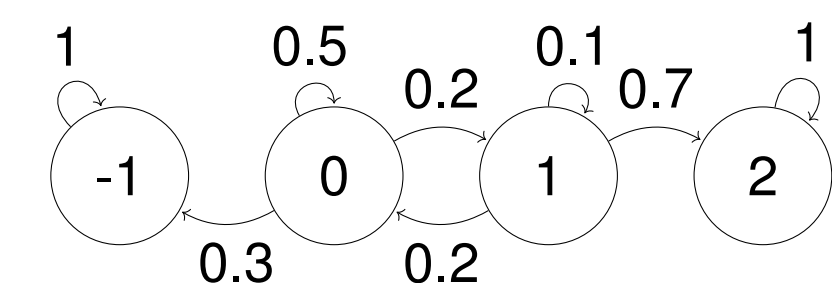


11. Dual Matrix

$$\text{Suppose } P = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0.7 & 0.3 & 0 & 0 \\ 0.2 & 0.6 & 0.2 & 0 \\ 0 & 0.7 & 0.3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$P^* = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.3 & 0.5 & 0.2 & 0 \\ 0 & 0.2 & 0.1 & 0.7 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Assume $P = [P_{i,j}]$ for i, j going from $0, 1, \dots, H$ is a $(H+1) \times (H+1)$ stochastic matrix. Suppose that the $(H+2) \times (H+2)$ matrix $P^* = [P^*_{i,j}]$ where:

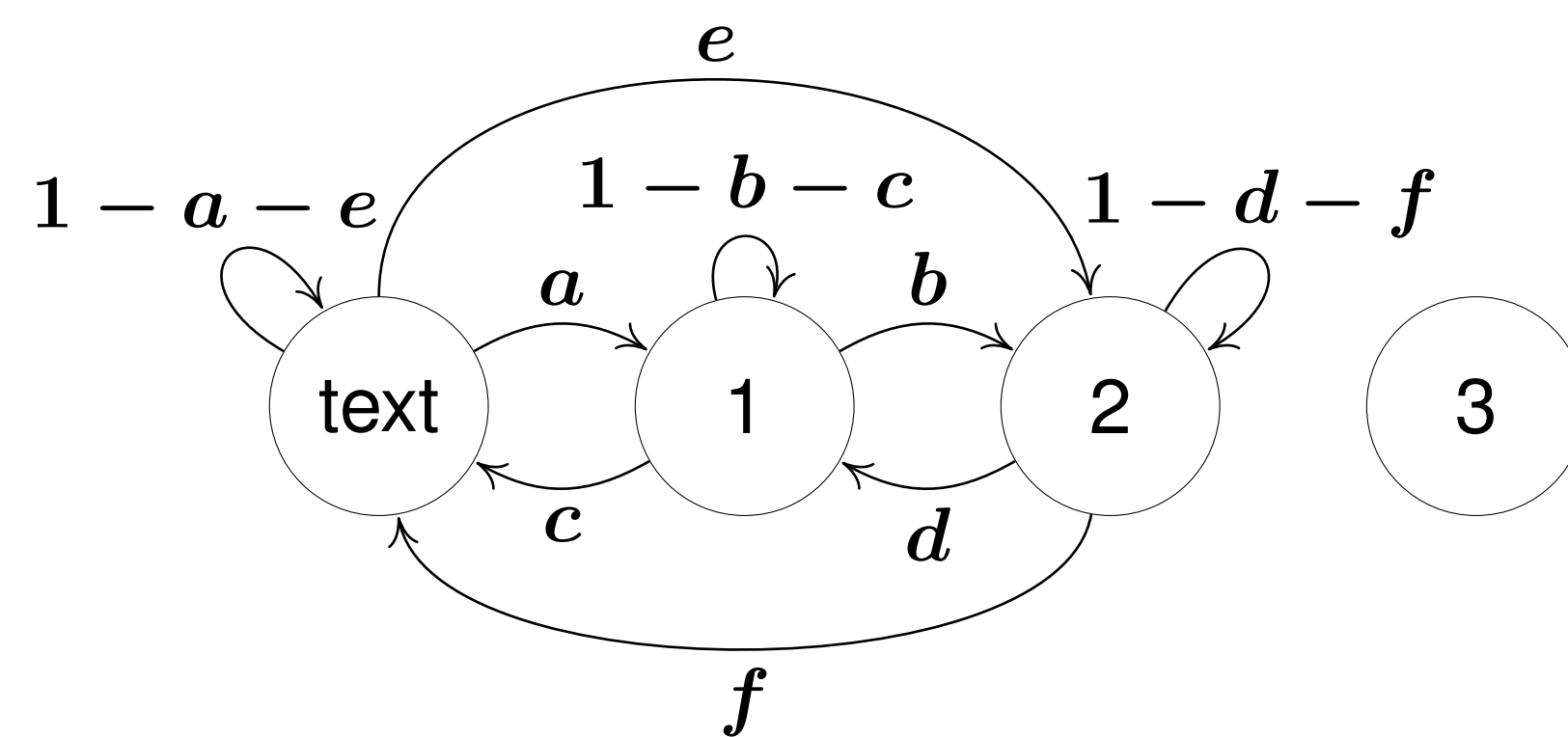
$$P^*_{i,j} = \sum_{s=i+1}^H [P_{j+1,s} - P_{j,s}]$$

is also a stochastic matrix under the following conventions:

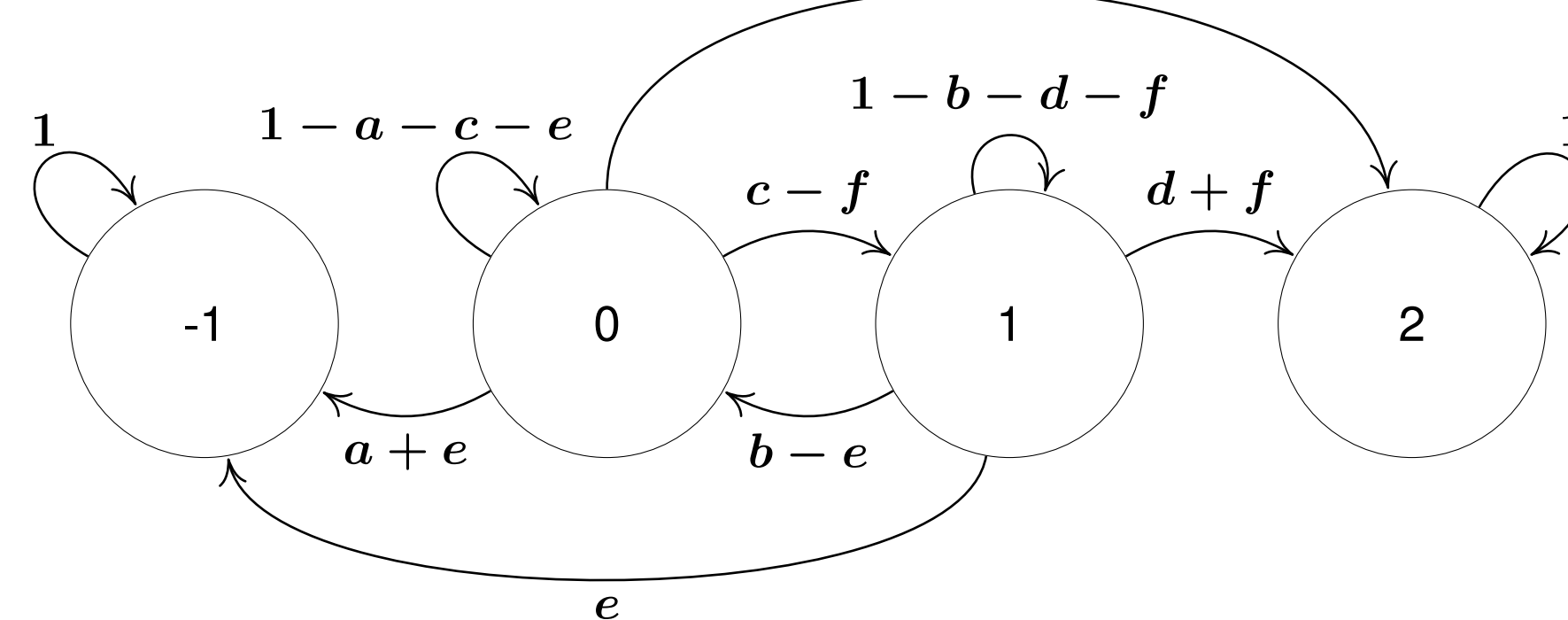
$$\begin{aligned} P^*_{-1,-1} &= 1 \\ P^*_{H,H} &= 1 \end{aligned}$$

Then P^* is called the **dual matrix** of P . **Note that P^* may not exist as a stochastic matrix. P and P^* have the same set of eigenvalues, see [4].**

12. Steady State Versus The Gambler's Ruin



$$P = \begin{bmatrix} 1-a-e & a & e \\ c & 1-b-c & b \\ f & d & 1-d-f \end{bmatrix}$$



$$P^* = \begin{bmatrix} 1 & 0 & 0 & 0 \\ a+e & 1-a-c-e & f+c & f \\ e & b-e & 1-b-d-f & d+f \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Wolfram Alpha finds a right eigenvector of P^T corresponding to eigenvalue 1 to be $\vec{v}_1$, which can be found to be

$$\vec{v}_1 = \begin{pmatrix} cd + bf + cf \\ ad + de + af \\ ab + be + ce \end{pmatrix}$$

The steady state distribution, $\vec{\pi}$, of the matrix P is equal to

$$\begin{aligned} \vec{\pi} &= (\pi(0), \pi(1), \pi(2)) \\ &= \left(\frac{A}{A+B+C}, \frac{B}{A+B+C}, \frac{C}{A+B+C} \right), \end{aligned}$$

where $A = cd + bf + cf$, $B = ad + de + af$, and $C = ab + be + ce$. The Gambler's Ruin probabilities within the dual matrix P^* are determined as follows:

$$\begin{aligned} P^*_{0,-1}^{*,(\infty)} &= \pi(1) + \pi(2) \\ P^*_{1,-1}^{*,(\infty)} &= \pi(2) \end{aligned}$$

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